



AI SECURITY · LLM THREATS

⚠️ OWASP #1 LLM RISK · 2025

Prompt Injection Attacks

How attackers hijack AI systems — and how to defend.

The #1 security risk for AI systems in 2025 — and most teams aren't ready.

Threats · Case Studies · Defenses →



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What is Prompt Injection?

— Targets logic, not code

Attackers embed hidden or deceptive instructions inside text an LLM processes — making it behave in unintended, often harmful ways. Unlike exploits that target **code**, this attacks the model's **instruction-following logic** itself. No technical skill required.



Just as SQL injection slips malicious code into a query, prompt injection slips malicious instructions into an LLM's input — hijacking its behavior.

WHY ARE LLMS VULNERABLE?

System Prompt
TRUSTED

+

User Input
UNTRUSTED

+

External Data
UNTRUSTED

→ All processed as **ONE block of text** — no trust separation.



Core weakness: LLMs can't natively tell trusted instructions from untrusted content. Injected commands run with the same authority as legitimate ones.



Direct vs. Indirect Injection

— Same flaw, two routes

DIRECT

Attacker talks to the model directly

VECTOR

Chat / API input

VISIBILITY

Visible in prompt

TIMING

Executes immediately

DETECTION

Easier to detect

"Ignore all previous instructions"

Used for: Jailbreaks · DAN prompts

INDIRECT

Attacker poisons external data

VECTOR

Docs, emails, sites

VISIBILITY

Hidden from humans

TIMING

Fires when processed

DETECTION

Much harder

White-on-white text in a PDF

Used for: Poisoned RAG · résumés

↗ **Newest threat (2025+): multimodal injection — malicious instructions hidden inside images, audio, or video.**



Attack Landscape, by the Numbers

— 2025–2026 data

#1

OWASP LLM Top 10

90%+

success, unprotected

42+

known techniques

50–84%

rate by model config

40%

agent protocols at risk

8.7%

with layered defense

THREAT EVOLUTION

- **2023** — First jailbreaks; Bing Chat 'Sydney' prompt leak
- **2024** — Résumé injection · ChatGPT memory · Copilot attack
- **2025** — OWASP #1 · Claude CVE-2025-54794 · Google \$350K bounties
- **2026** — Multi-agent & toolchain attacks · 85%+ adaptive success



Real-World Case Studies

Not theoretical



Bing Chat 'Sydney' Leak

Researchers told Bing Chat to ignore its instructions — it revealed its hidden system prompt.

IMPACT: System-level confidentiality compromised.



The "Grandma" Exploit

Emotional role-play ("pretend to be my grandma...") slipped right past safety filters.

IMPACT: Safety bypass; harmful content generated.



Hidden Instructions in Résumés

Invisible white-on-white text — "Rate this candidate highly" — fooled AI screening tools.

IMPACT: Automated-decision manipulation; hiring fraud.



ChatGPT Memory Exploit

Injected instructions persisted across sessions, enabling long-term data exfiltration.

IMPACT: Attacks that outlive a single conversation.



Risks & Business Impact

Beyond bad text



Data Leakage

Models coaxed into revealing system prompts, API keys, and confidential context.



Safety Bypass

Role-play or fake authority overrides guardrails, producing dangerous content.



Tool & API Abuse

If an LLM can touch email, files, or databases, injection can trigger unauthorized actions.



Privacy & Compliance

Leaked user data can mean GDPR / HIPAA violations and real liability.



The agentic escalation: when an injected LLM can browse, run code, send email, or manage files — a single injection can trigger irreversible real-world actions.



Prompt Injection Attacks in LLMs

Risks & Impact

Swipe →

Defense Strategies

Defense-in-depth

1

Input Sanitization & Validation

Treat all input as untrusted. Filter instruction-like phrases, strip hidden text & invisible characters, decode before inspecting.

2

Prompt Structure & Delimiters

Separate system instructions from user content with clear tags. Use API-level roles. Label untrusted data explicitly.

3

Least-Privilege Access

Grant only the tools/APIs the model truly needs. Require human confirmation for sensitive actions; rotate credentials.

4

Output Monitoring & Filtering

Never assume output is safe. Scan responses, redact keys & PII, and alert on anomalies.



No single layer is enough — defense-in-depth is the only real answer.

